

Media Access Control Sub Layer

Lecture#33-38



Dr. Sanjeev Patel

Asst Professor, CSE Dept.

NIT Rourkela

Outline

- MAC Protocol
- Contention-free and contention-based protocol
- Random access protocol
- Aloha Protocol
- CSMA protocol
- Ethernet
- CSMA/CD
- Polling
- Reservation

Responsibility of MAC Layer

❑ Responsibility of MAC Layer

- decide when a node accesses a shared medium
- resolve any potential conflicts between competing nodes
- correct communication errors occurring at the physical layer
- perform other activities such as framing, addressing, and flow control

❑ Second layer of the OSI reference model (data link layer)

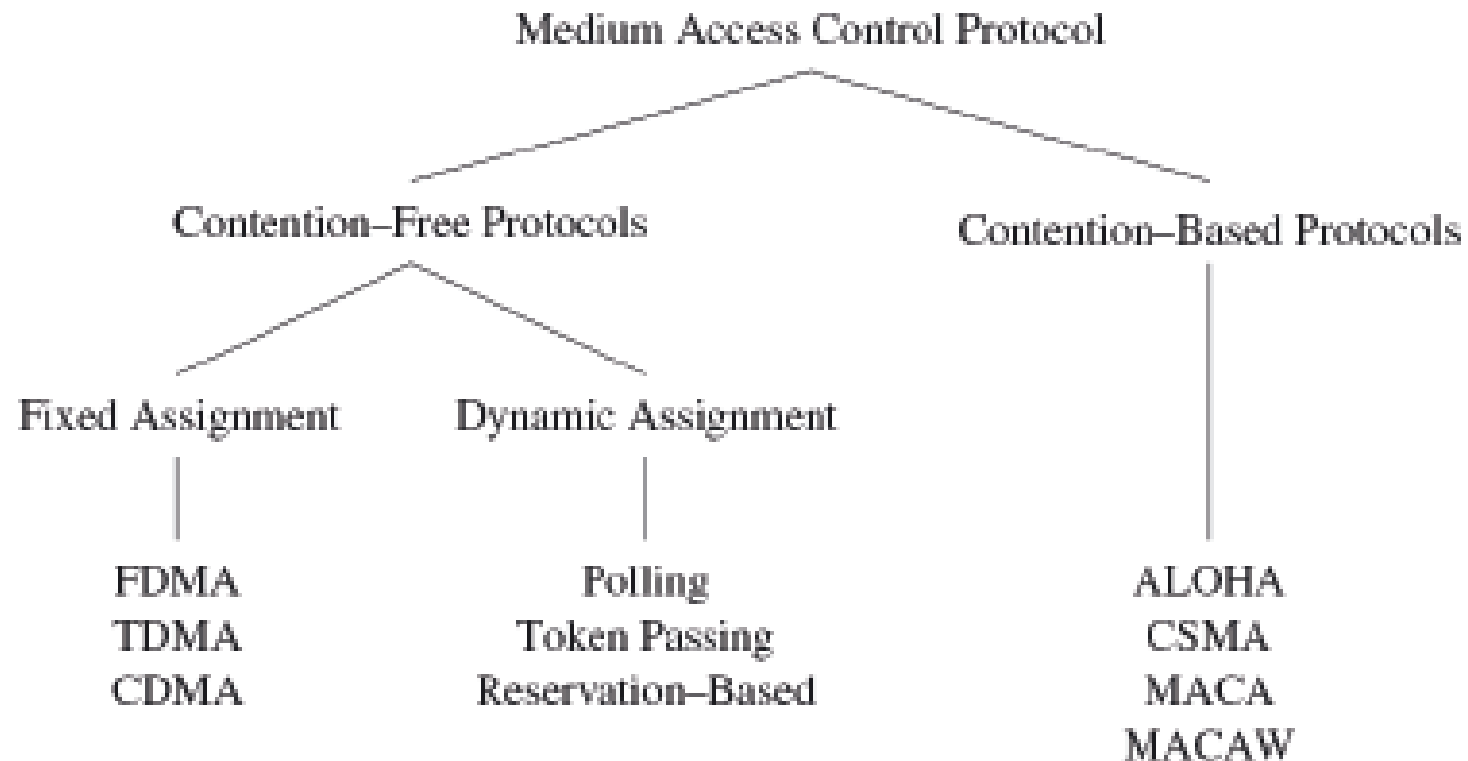
The Channel Allocation Problem [2]

- How to allocate a single broadcast channel among competing users
- Static Channel Allocation in LANs and MANs
 - The traditional way of allocating a single channel, such as a telephone trunk, among multiple competing users is Frequency Division Multiplexing (FDM).
 - If there are N users, the bandwidth is divided into N equal-sized portions, each user being assigned one portion.
- Dynamic Channel Allocation in LANs and MANs
 - Random allocation of the channel under following assumptions

Dynamic Channel Allocation in LANs and MANs

1. Station Model- The model consists of N independent stations
2. Single Channel Assumption- A single channel is available for all communication
3. Collision Assumption- If two frames are transmitted simultaneously, they overlap in time and the resulting signal is garbled. This event is called a collision.
4. (a) Continuous Time- Frame transmission can begin at any instant
(b) Slotted Time.
5. (a) Carrier Sense- If the channel is sensed as busy, no station will attempt to use it until it goes idle.
(b) No Carrier Sense.

MAC Protocols



MAC Protocols: a taxonomy

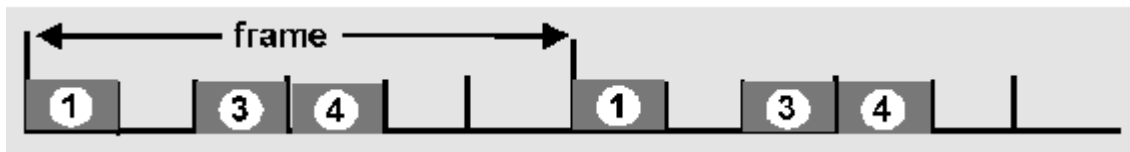
Three broad classes:

- **Channel Partitioning**
 - divide channel into smaller “pieces” (time slots, frequency, code)
 - allocate piece to node for exclusive use
- **Random Access**
 - channel not divided, allow collisions
 - “recover” from collisions
- **“Taking turns”**
 - Nodes take turns, but nodes with more to send can take longer turns

Channel Partitioning MAC protocols: TDMA

TDMA: time division multiple access

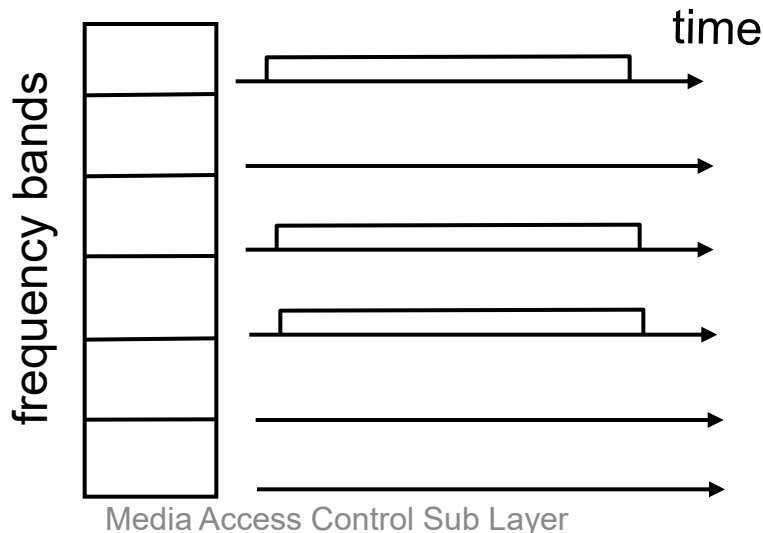
- access to channel in "rounds"
- each station gets fixed length slot (length = pkt trans time) in each round
- unused slots go idle
- example: 6-station LAN, 1,3,4 have pkt, slots 2,5,6 idle



Channel Partitioning MAC protocols: FDMA

FDMA: frequency division multiple access

- channel spectrum divided into frequency bands
- each station assigned fixed frequency band
- unused transmission time in frequency bands go idle
- example: 6-station LAN, 1,3,4 have pkt, frequency bands 2,5,6 idle

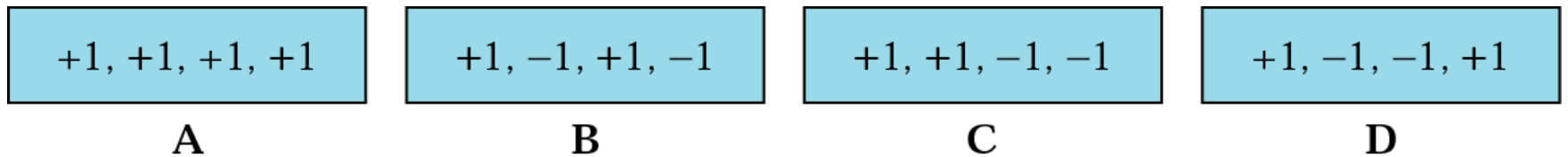


CDMA

- One channel occupies entire bandwidth of the link(differs from FDMA)
- All stations can send data simultaneously(differs from TDMA)
- One channel carries all transmission simultaneously
- It is based on coding theory
- Each stations is assigned a code , which is a sequence of numbers called chips

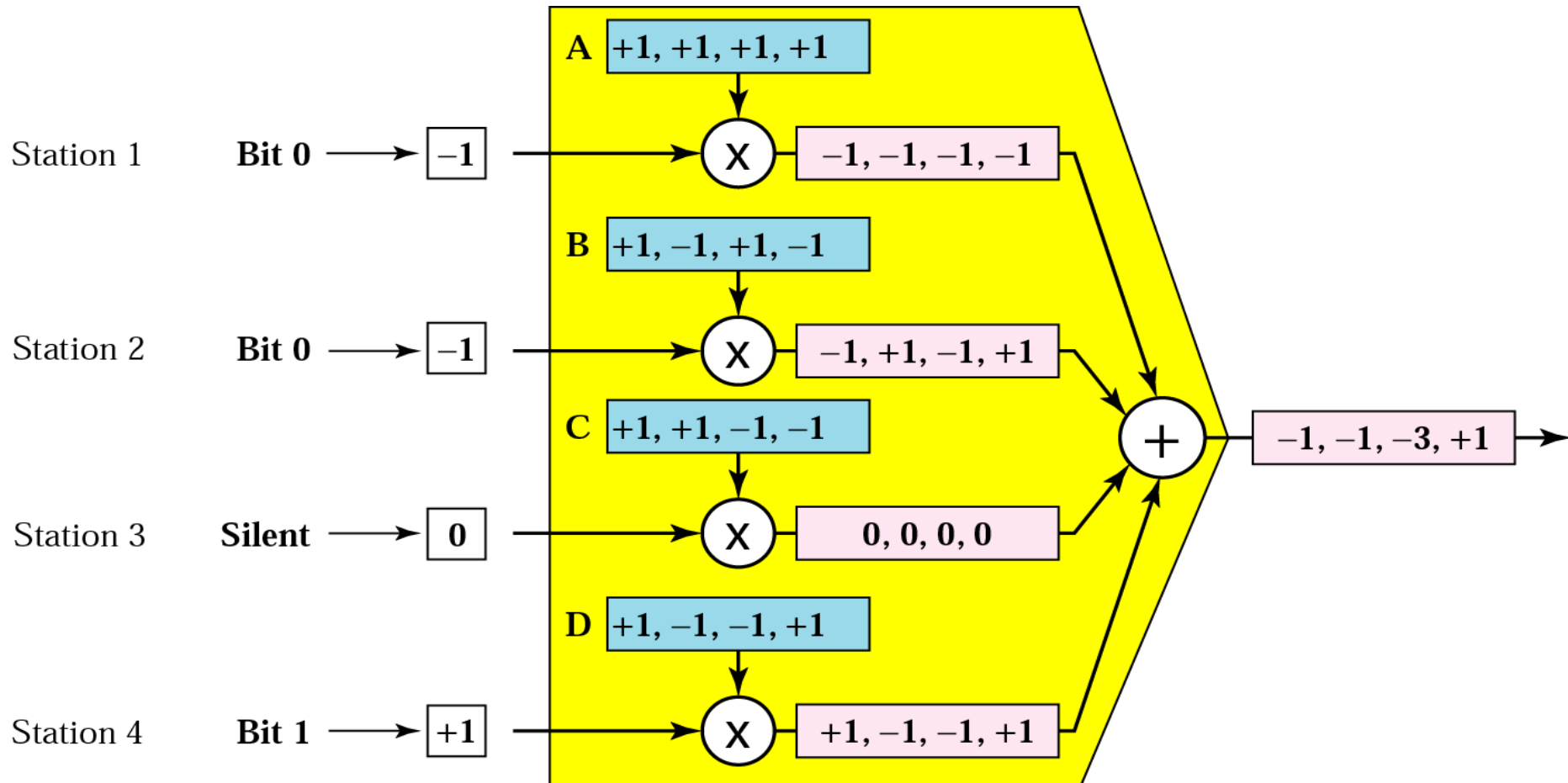
Rules for encoding

- Suppose we have a sequence of four chips corresponding to each station

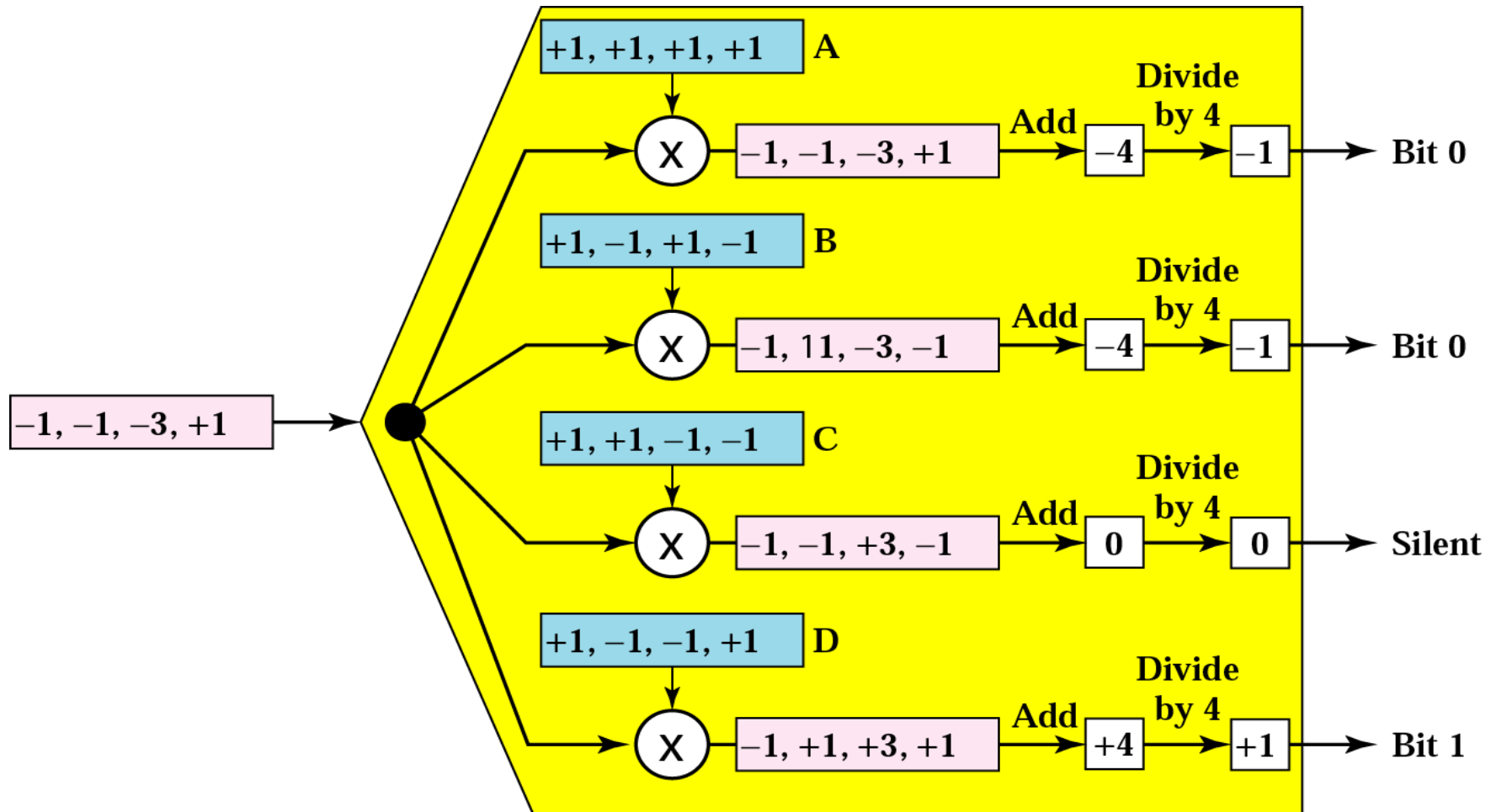


Data bit 0 $\longrightarrow$ -1 **Data bit 1** $\longrightarrow$ $+1$ **Silence** $\longrightarrow$ 0

CDMA multiplexer [5]



CDMA demultiplexer [5]



Random Access Protocols

- When node has packet to send
 - transmit at full channel data rate R .
 - no *a priori* coordination among nodes
- two or more transmitting nodes → “collision”,
- random access MAC protocol specifies:
 - how to detect collisions
 - how to recover from collisions (e.g., via delayed retransmissions)
- Examples of random access MAC protocols:
 - slotted ALOHA
 - ALOHA
 - CSMA, CSMA/CD, CSMA/CA

Slotted ALOHA

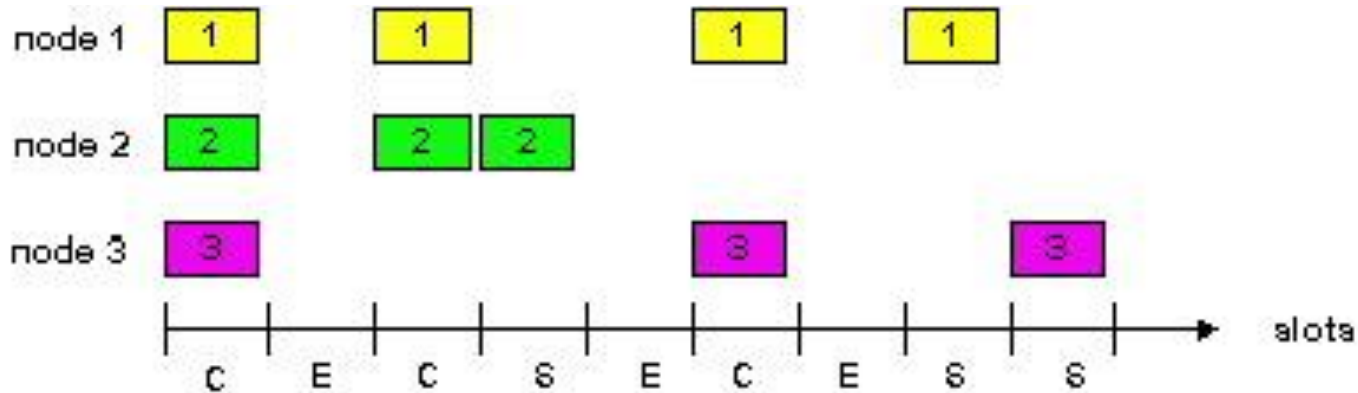
Assumptions

- all frames same size
- time is divided into equal size slots, time to transmit 1 frame
- nodes start to transmit frames only at beginning of slots
- nodes are synchronized
- if 2 or more nodes transmit in slot, all nodes detect collision

Operation

- when node obtains fresh frame, it transmits in next slot
- no collision, node can send new frame in next slot
- if collision, node retransmits frame in each subsequent slot with prob. p until success

Slotted ALOHA [6]



Pros

- single active node can continuously transmit at full rate of channel
- highly decentralized: only slots in nodes need to be in sync
- simple

Cons

- collisions, wasting slots
- idle slots
- nodes may be able to detect collision in less than time to transmit packet
- clock synchronization

Slotted Aloha efficiency

Efficiency is the long-run fraction of successful slots when there are many nodes, each with many frames to send

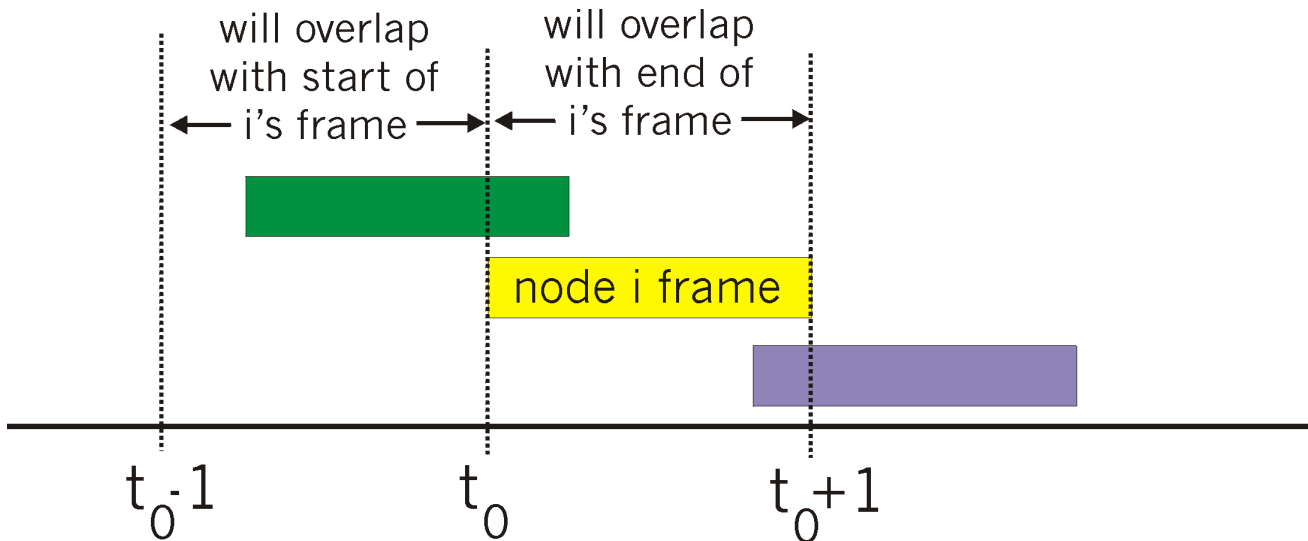
- Suppose N nodes with many frames to send, each transmits in slot with probability p
- prob that node 1 has success in a slot $= p(1-p)^{N-1}$
- prob that any node has a success $= Np(1-p)^{N-1}$

- For max efficiency with N nodes, find p^* that maximizes $Np(1-p)^{N-1}$
- For many nodes, take limit of $Np^*(1-p^*)^{N-1}$ as N goes to infinity, gives $1/e = .37$

At best: channel used for useful transmissions 37% of time!

Pure (unslotted) ALOHA

- unslotted Aloha: simpler, no synchronization
- when frame first arrives
 - transmit immediately
- collision probability increases:
 - frame sent at t_0 collides with other frames sent in $[t_0-1, t_0+1]$



Pure Aloha efficiency [6]

$P(\text{success by given node}) = P(\text{node transmits}) \cdot$

$P(\text{no other node transmits in } [t_0-1, t_0] \cdot$

$P(\text{no other node transmits in } [t_0, t_0+1])$

$$= p \cdot (1-p)^{N-1} \cdot (1-p)^{N-1}$$

$$= p \cdot (1-p)^{2(N-1)}$$

... choosing optimum p and then letting $n \rightarrow \infty$...

Even worse ! $= 1/(2e) = .18$

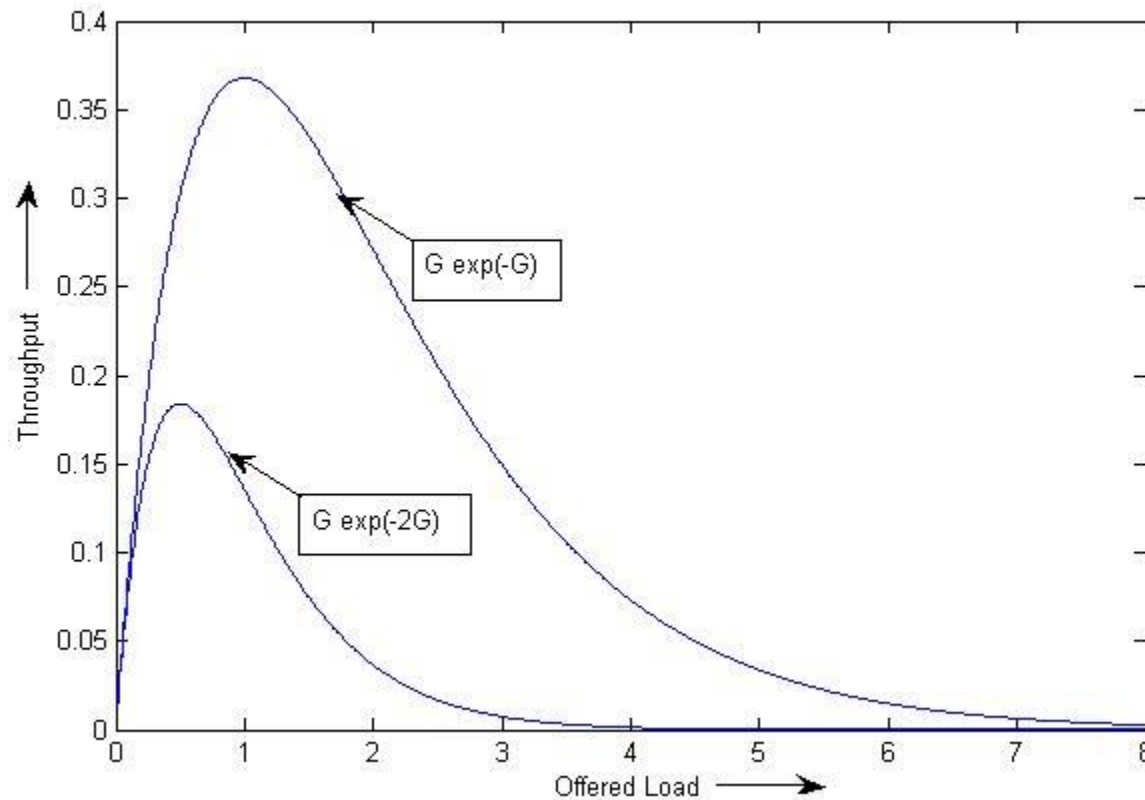
Alternative Method

- Offered load is the average number of frames transmitted per unit time.
- Arrival of frames or packets follow the Poisson process.
- Therefore, probability of k arrival of frames in t time : $\{(\lambda t)^k \exp(-\lambda t)\} / k!$
- Assume $t = 1$ frame transmission unit and $\lambda = G$,
 - then $\text{Prob}[k \text{ frames}] = \{G^k \exp(-G)\} / k!$

Alternative Method

- The probability that it will avoid a collision is e^{-G} , that is the probability that all the other users are silent in that slot
- Probability of success= $\text{Prob}[k=0] = e^{-G}$
- Probability of (collision +empty)= $\text{Prob}[k>0] = 1-e^{-G}$
- Efficiency of slotted Aloha is calculated as:
 - Efficiency is offered load X $\text{Prob}[\text{success}] = G * e^{-G}$
 - At $G=1$, empty slot=37%, success=37%, collision=26%.
- For pure Aloha, vulnerable time is 2 slots
- Efficiency of pure Aloha is:
offered load X $\text{Prob}[\text{success}] = G * e^{-2G}$

Throughput Comparison



Carrier Sense Multiple Access

- To minimize the no. of collision or maximize the performance, we can use the concept of sensing the medium
- It is based on the technique listen before talk or sense before transmit
- But it can not eliminate all the collision
- It is able to reduce the collision and improve the throughput

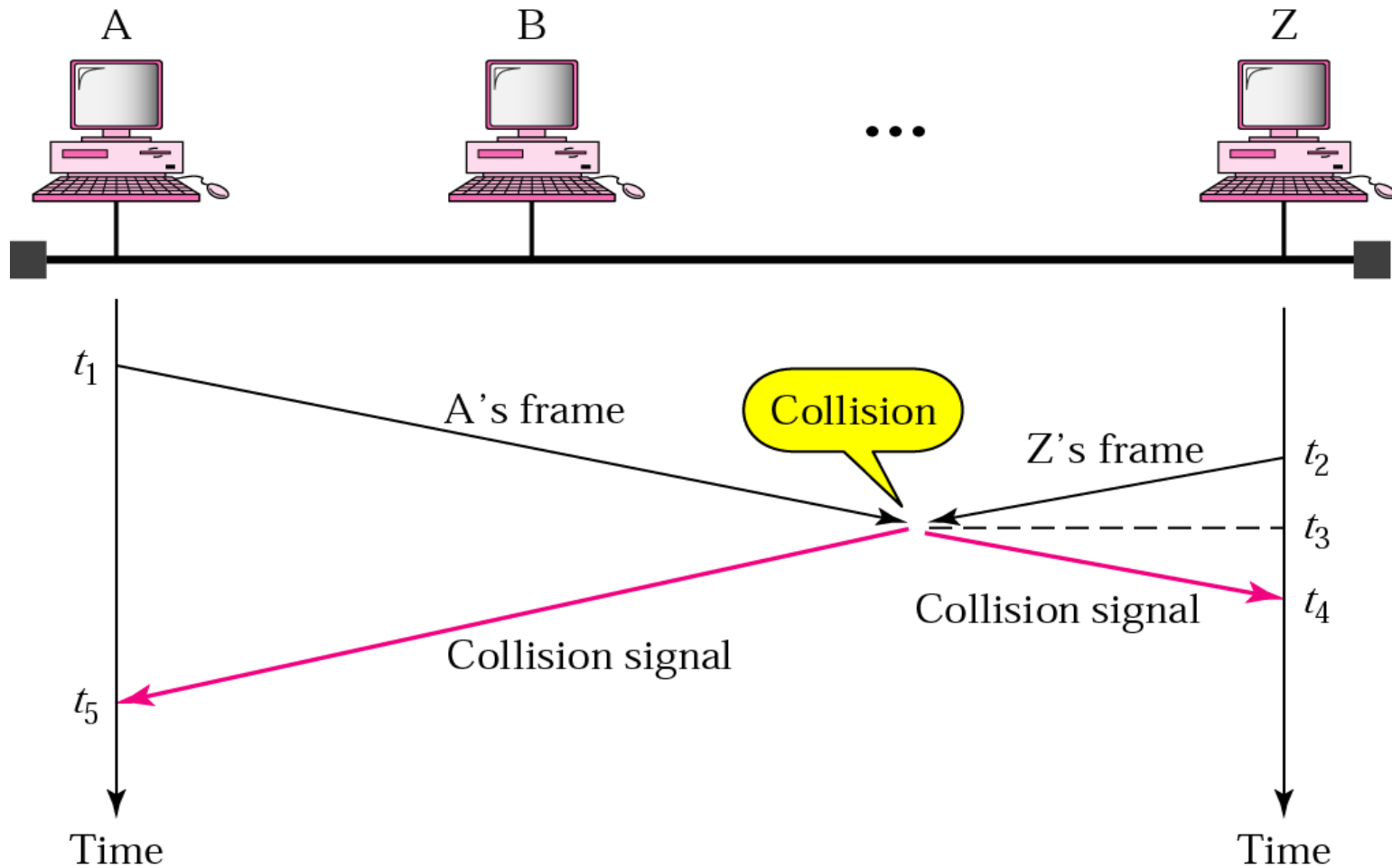
CSMA (Carrier Sense Multiple Access)

CSMA: listen before transmit:

If channel sensed idle: transmit entire frame

- If channel sensed busy, defer transmission
- Human analogy: don't interrupt others!

Collision in CSMA [5]



Persistence Technique

- This is technique defined that what should we do when medium is busy
- **Non-persistence:**
 1. If the medium is idle ,transmit; otherwise , go to step 2
 2. If the medium is busy , wait an amount of time drawn from a probability distribution(retransmission delay) and repeat step 1

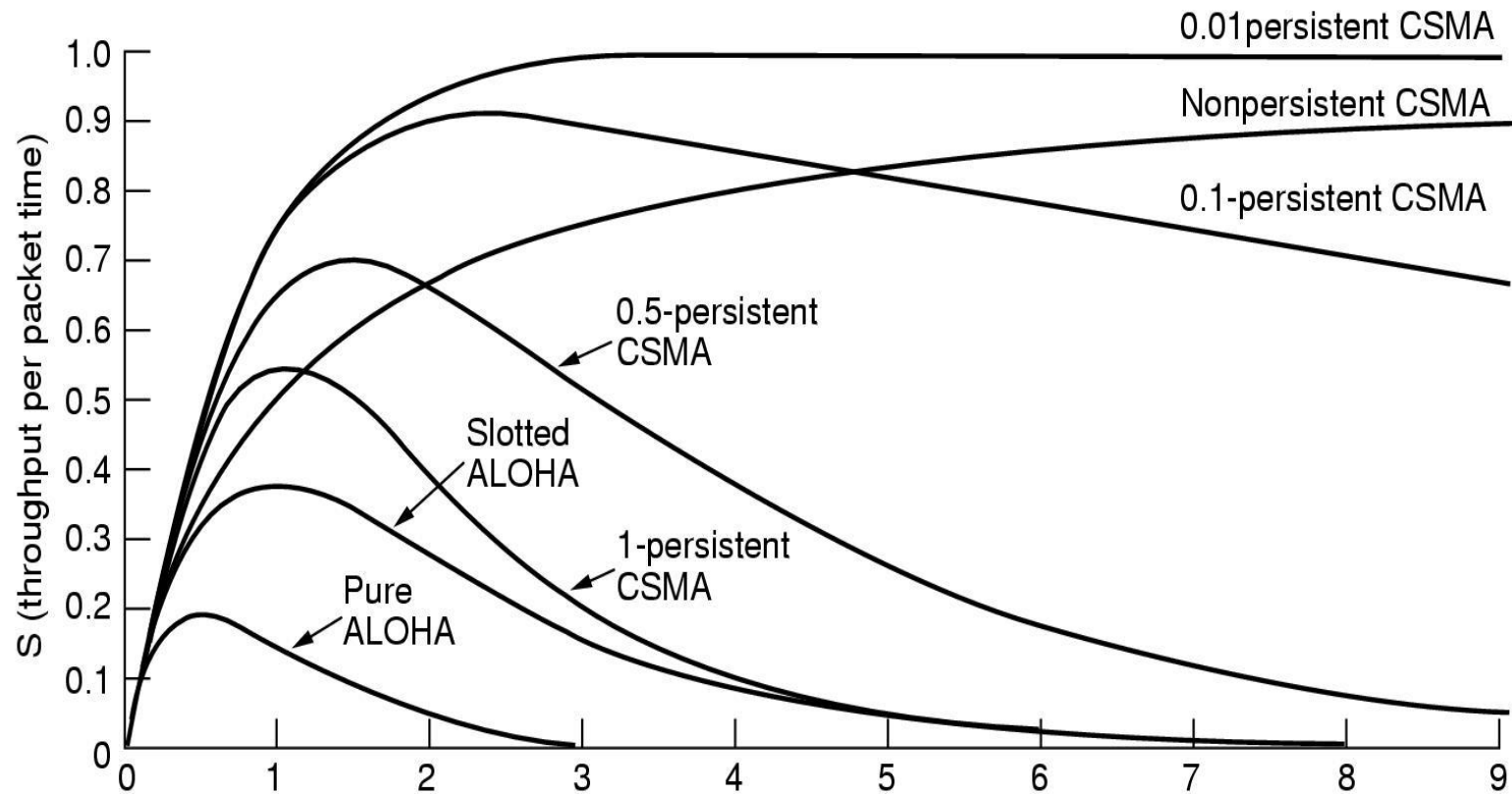
1-Persistence

- Algorithm:
 1. If the medium is idle ,transmit; otherwise , go to step 2
 2. If the medium is busy , continue to listen until the channel is sensed idle; then transmit immediately
- Still, there is a chance of collision, when two stations senses the medium simultaneously

p-Persistence

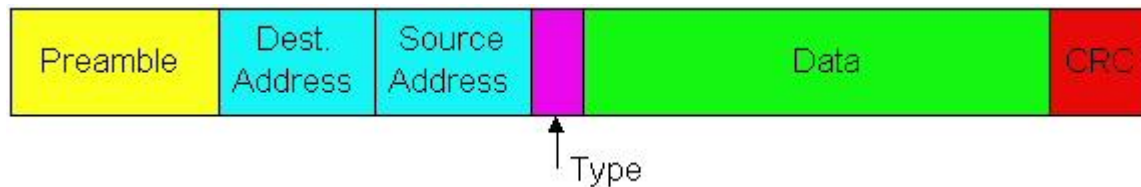
- Algorithm:
 1. If the medium is idle ,transmit with probability p , and delay one time unit with probability $(1-p)$ (The time unit is typically equal to the maximum propagation delay)
 2. If the medium is busy, continue to listen until the channel is idle and repeat step 1
 3. If the transmission is delayed one time unit, repeat step 1
- It shows the highest performance among the all persistence technique

Throughput Comparison [2]



Ethernet Frame Structure

Sending adapter encapsulates IP datagram (or other network layer protocol packet) in **Ethernet frame**



Preamble:

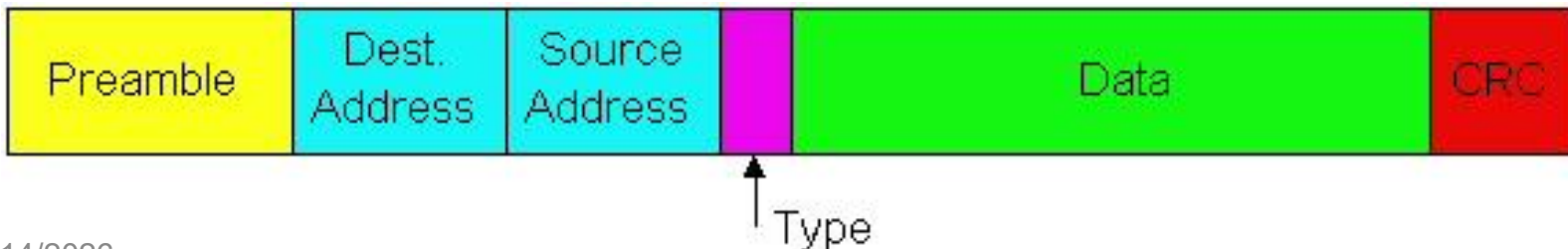
- 7 bytes and each byte has the pattern 10101010 followed by one byte with pattern 10101011
- used to synchronize receiver, sender clock rates

Data:

- 46-1500 bytes

Ethernet Frame Structure [6]

- **Addresses:** 6 bytes
 - if adapter receives frame with matching destination address, or with broadcast address (eg ARP packet), it passes data in frame to net-layer protocol
 - otherwise, adapter discards frame
- **Type: 2 bytes** indicates the higher layer protocol (mostly IP but others may be supported)
 - permits Ethernet to multiplex network-layer protocols
 - It also indicated length of data in IEEE 802.3
- **CRC:** checked at receiver, if error is detected, the frame is simply dropped



Bit Stuffing

- Ethernet start and end flag has a pattern 01111110
- Delimit frame at both ends 01111110
- May close one frame and open another
- Receiver hunts for flag sequence to synchronize
- Bit stuffing used to avoid confusion with data containing 01111110
 - 0 inserted after every sequence of five 1s
 - If receiver detects five 1s it checks next bit
 - If 0, it is deleted
 - If 1 and seventh bit is 0, accept as flag
 - If sixth and seventh bits 1, sender is indicating abort

Bit Stuffing

- Example with possible errors

Original Pattern:

111111111111110111111101111110

After bit-stuffing

1111101111110110111111010111111010

(a) Example



(b) An inverted bit splits a frame in two

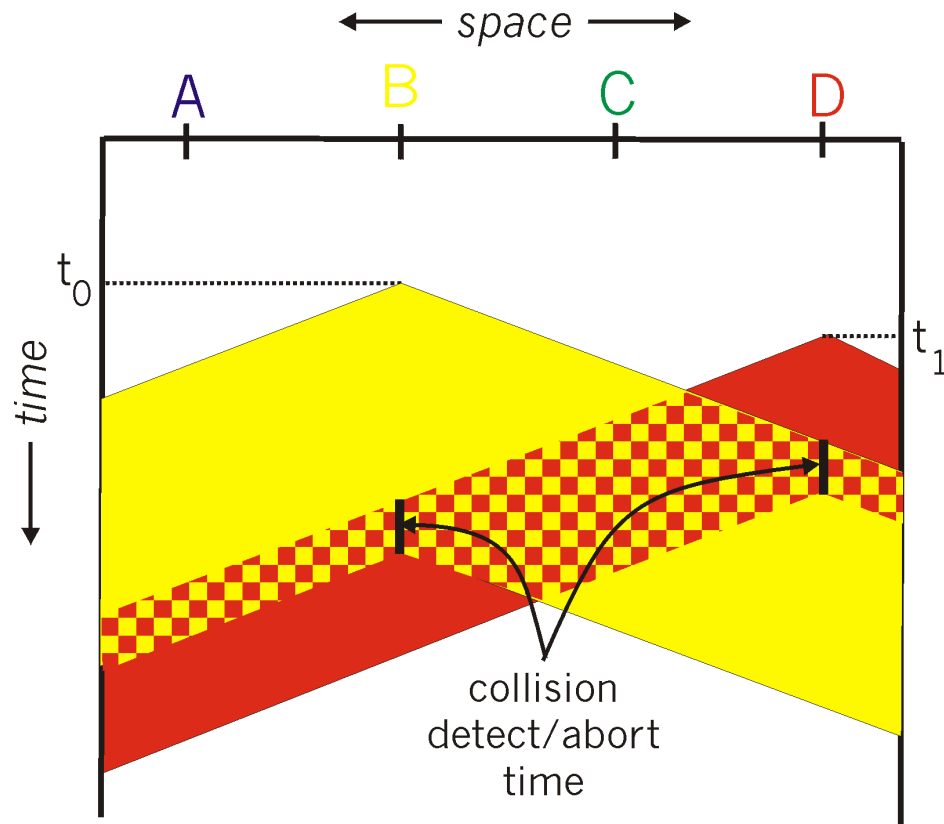


(c) An inverted bit merges two frames

CSMA/CD (Collision Detection)

- CSMA/CD:** carrier sensing, deferral as in CSMA
- collisions *detected* within short time
 - colliding transmissions aborted, reducing channel wastage
 - collision detection:
 - easy in wired LANs: measure signal strengths, compare transmitted, received signals
 - difficult in wireless LANs: receiver shut off while transmitting

CSMA/CD collision detection



Ethernet uses CSMA/CD

- No slots
- Carrier sense-
 - Adapter doesn't transmit if it senses that some other adapter is transmitting
- Collision detection –
 - Transmitting adapter aborts when it senses that another adapter is transmitting
- Random access –
 - Before attempting a retransmission, adapter waits a random time

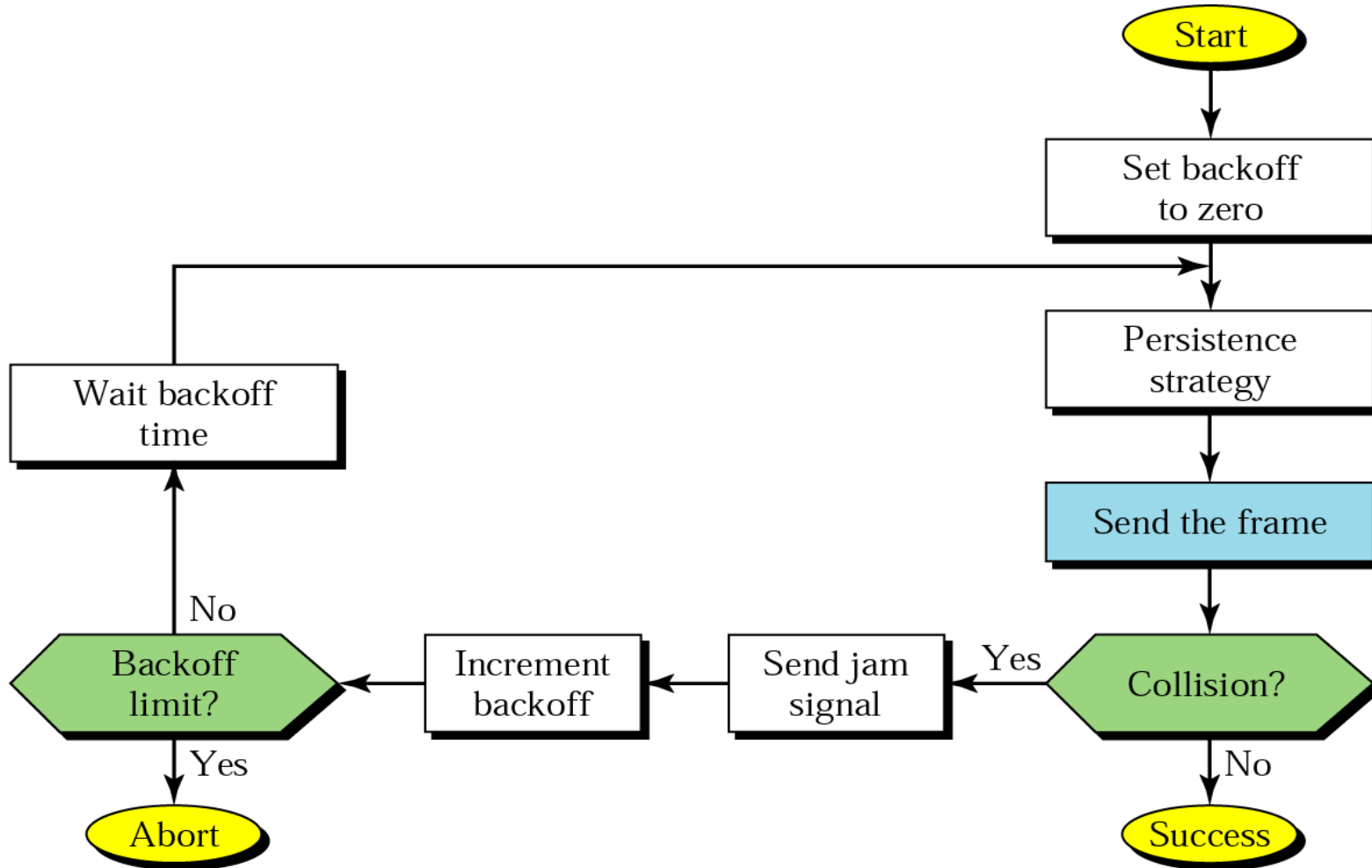
Contd...

- **Back-off-**
 - If there is a collision, the frame needs to be sent again.
 - To reduce the probability of collision, the station waits up to the back-off limit and increment by one corresponding to each collision
 - If station does not hears a collision until the whole frame has been sent, the transmission is successful.
- **Jamming signal-**
 - If the station hear the collision, it sends a jam signal to all to alert them that a collision has occurred;
 - all stations discard the part of the frame received.
 - The station then increments the value of the back-off parameter by 1

Ethernet CSMA/CD algorithm [6]

1. Adaptor receives datagram from net layer & creates frame
2. If adapter senses channel idle for 96 bit times, it starts to transmit frame. If it senses channel busy, waits until channel idle plus 96 bit times and then transmits
3. If adapter transmits entire frame without detecting another transmission, the adapter is done with frame !
4. If adapter detects another transmission while transmitting, aborts and sends jam signal of 48 bit times.
5. After aborting, adapter enters **exponential backoff**: after the m th collision, adapter chooses a K at random from $\{0, 1, 2, \dots, 2^m - 1\}$. Adapter waits $K \cdot 512$ bit times and returns to Step 2

CSMA/CD procedure [5]



- Truncated Binary Exponential Backoff

- Attempts := 0;
- Attempt Limit is 16 and Backoff Limit 10.

While

{ (Frame NOT Transmitted Successfully)
AND (Attempts < Attempt Limit) }

K := Min (Attempts, Backoff Limit)

R := Random (0, $2^K - 1$)

delay := R * SlotTime;

Wait (delay)

Increment Attempts;

End While

- This algorithm yields a "**statistical fairness**" in the sharing of bandwidth.

CSMA/CD efficiency

- T_{prop} = max prop between 2 nodes in LAN
- t_{trans} = time to transmit max-size frame

$$\text{efficiency} = \frac{1}{1 + 5t_{\text{prop}} / t_{\text{trans}}}$$

- Efficiency goes to 1 as t_{prop} goes to 0
- Goes to 1 as t_{trans} goes to infinity
- Much better than ALOHA, but still decentralized, simple, and cheap

“Taking Turns” MAC protocols

channel partitioning MAC protocols:

- share channel efficiently and fairly at high load
- inefficient at low load: delay in channel access, $1/N$ bandwidth allocated even if only 1 active node!

Random access MAC protocols

- efficient at low load: single node can fully utilize channel
- high load: collision overhead

“taking turns” protocols (Controlled-access Protocol)

look for best of both worlds!

Controlled-access Protocols

- In this station consult one another to find which station has right to send
- The asynchronous approach can be further subdivided into many categories:
- Round robin and Reservation: Centralization can be handled by
 - Polling
 - Token Passing

Round Robin

- With round robin, each station in turn is given the opportunity to transmit.
- During that opportunity, the station may decline to transmit or may transmit subject to a specified upper bound
- Usually expressed as a maximum amount of data transmitted or time for this opportunity.
- When it is finished, relinquishes its turn, and the right to transmit passes to the next station in logical sequence.
- Control of sequence may be centralized or distributed. Polling is an example of a centralized technique

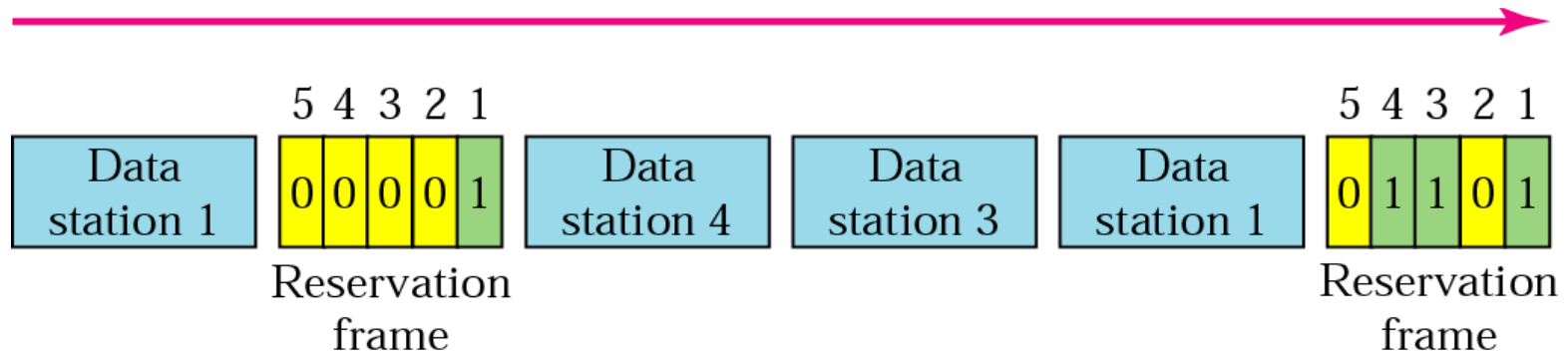
Reservation

- A station needs to make a reservation before sending data
- Time is divided into intervals
- A reservation frame precedes the data frames sent in that interval
- If there are N stations in the system, there are exactly N reservation minislots in the reservation frame
- Each minislot belongs to a station and each station makes a reservation in its own minislot

Contd.

- For stream traffic, reservation techniques are well suited.
- In this techniques, time on the medium is divided into slots, much as with synchronous TDM.
- A station wishing to transmit reserves future slots for an extended or even an indefinite period.
- Again, reservations may be made in a centralized or distributed fashion.

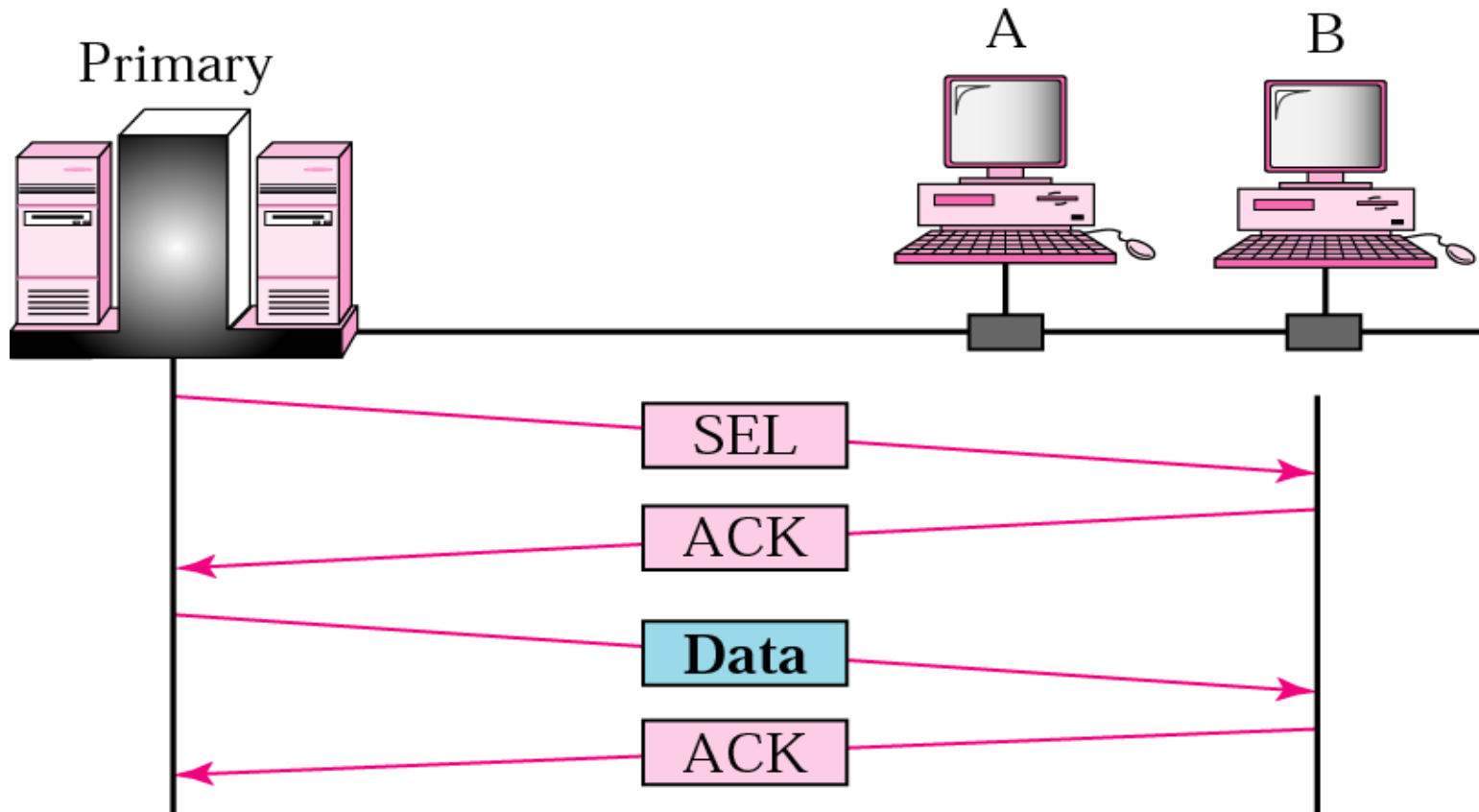
Reservation access method [5]



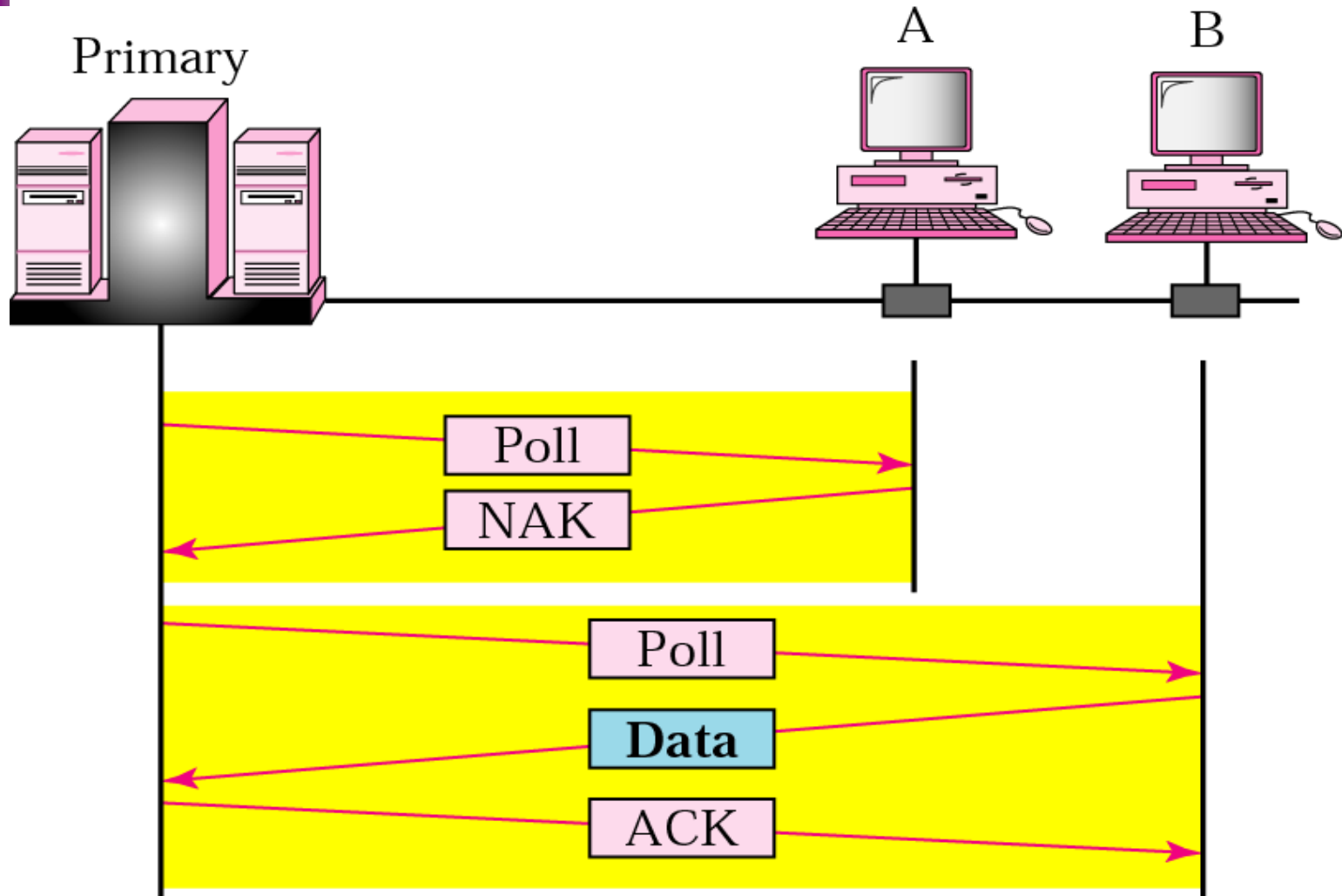
Polling

- It involves two type of devices
- Primary Station: rule maker
- Secondary Station: follow the instructions made by primary station
- Two operations are involved

Select [5]



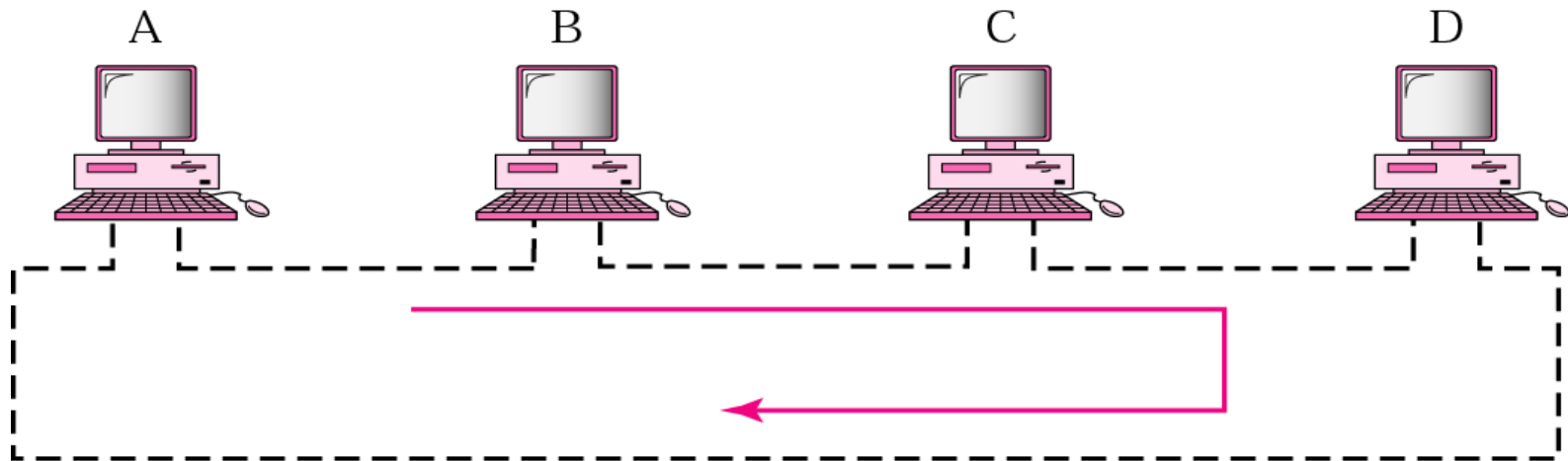
Poll [5]



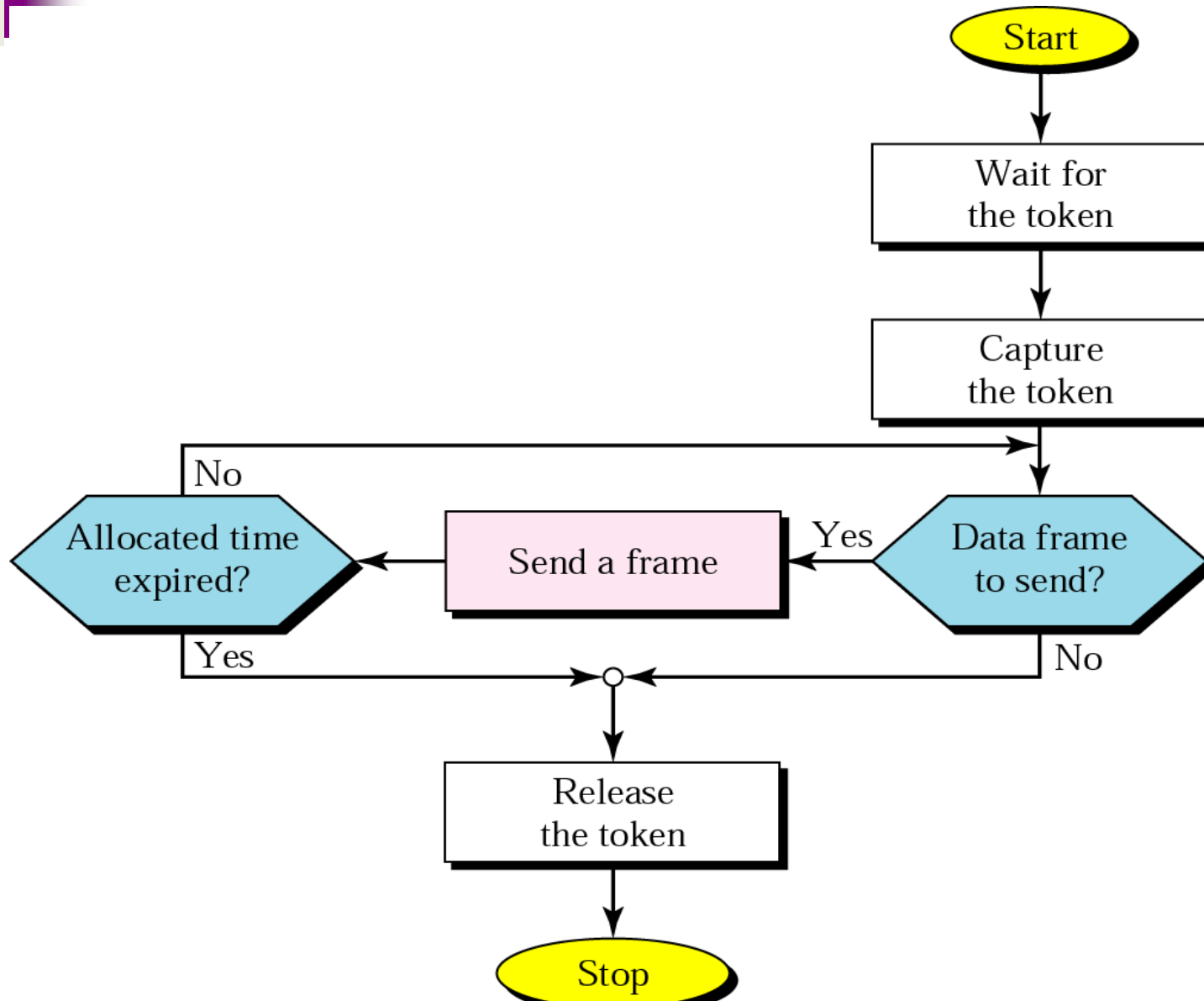
Token Passing

- A token circulates around the ring, if no data are being sent
- The station captures the token and sends one or more frames
- It will keep capture till they have frame or allocated time has not expired
- Finally release the token after used

Token-passing network [5]



Token-passing procedure [5]



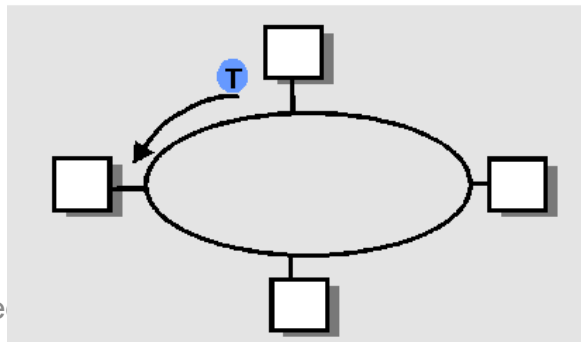
“Taking Turns” MAC protocols (Controlled Access) [6]

Polling:

- master node “invites” slave nodes to transmit in turn
- concerns:
 - polling overhead
 - latency
 - single point of failure (master)

Token passing:

- ❑ control **token** passed from one node to next sequentially.
- ❑ token message
- ❑ concerns:
 - token overhead
 - latency
 - single point of failure (token)



Repeater [9]

- ❏ “A repeater receives a signal, regenerates it, and passes it on.
- ❏ It can regenerate and retiming network signals at the bit level to allow them to travel a longer distance on the media.
- ❏ It operates at Physical Layer of OSI
- ❏ The Four Repeater Rule for 10-Mbps Ethernet should be used as a standard when extending LAN segments.
- ❏ This rule states that no more than four repeaters can be used between hosts on a LAN.
- ❏ This rule is used to limit latency added to frame travel by each repeater.”

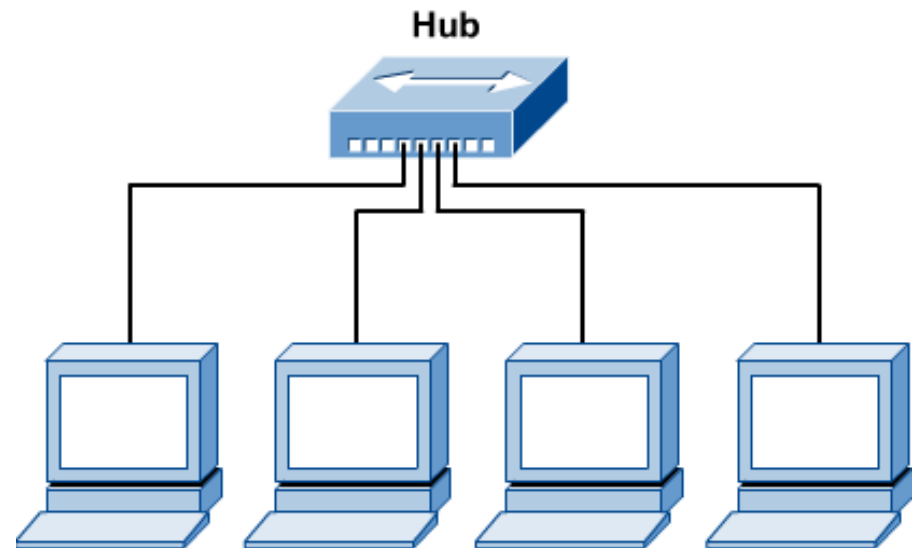
Hub [9]

“Hubs are used to connect multiple nodes to a single physical device, which connects to the network.

Hubs are actually multiport repeaters.

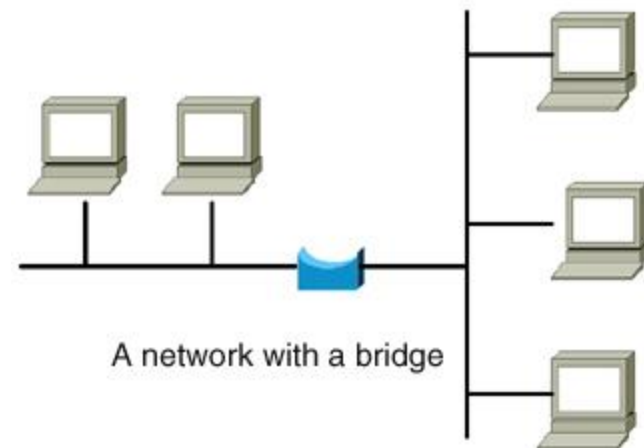
Using a hub changes the network topology from a linear bus, to a star.

With hubs, data arriving over the cables to a hub port is electrically repeated on all the other ports connected to the same network segment, except for the port on which the data was sent. “



Bridge [9]

- ❏ “Bridges are used to logically separate network segments within the same network.
- ❏ They operate at the OSI data link layer (Layer 2) and are independent of higher-layer protocols.
- ❏ The function of the bridge is to make intelligent decisions about whether or not to pass signals on to the next segment of a network.
- ❏ When a bridge receives a frame on the network, the destination MAC address is looked up in the bridge table to determine whether to filter, flood, or copy the frame onto another segment. Broadcast Packets are forwarded”



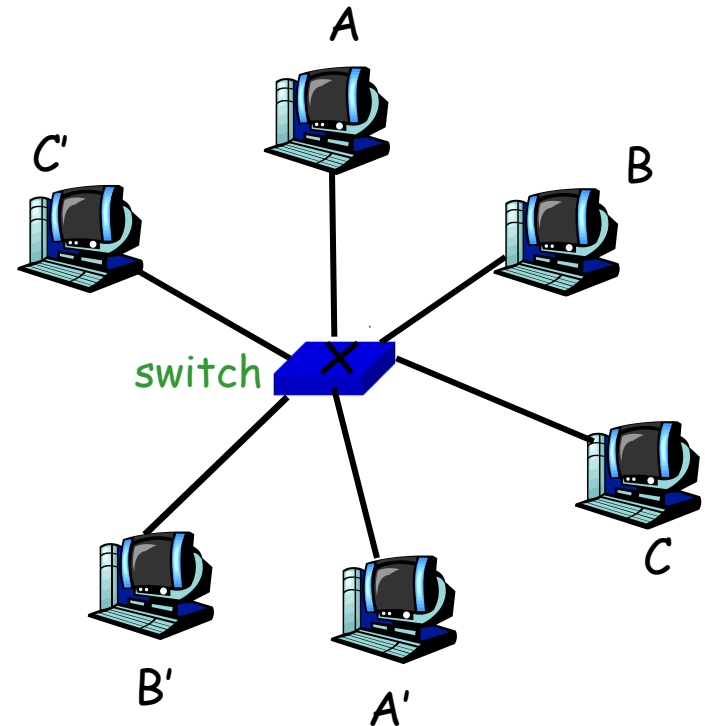
Switch [9]



- ❏ “Switches are Multiport Bridges.
- ❏ Switches provide a unique network segment on each port, thereby separating collision domains.
- ❏ Today, network designers are replacing hubs in their wiring closets with switches to increase their network performance and bandwidth while protecting their existing wiring investments.
- ❏ Like bridges, switches learn certain information about the data packets that are received from various computers on the network.
- ❏ Switches use this information to build forwarding tables to determine the destination of data being sent by one computer to another computer on the network. “

Switches: Dedicated Access [9]

- ❏ “Hosts have direct connection to switch
- ❏ Full Duplex: No collisions
- ❏ Switching: A-to-A' and B-to-B' simultaneously, no collisions
- ❏ Switches can be cascaded to expand the network”



Router and Gateway

- **Router**

- It is used to determine the routing decision of the incoming packet using routing table stored in router
- It is responsible for receiving, analyzing and forwarding the data packets to other networks and operated at network layer

- **Gateways**

- It regulates traffic between two dissimilar networks, while **routers** regulate traffic between similar networks.
- It is responsible for the conversion of one protocol into the other where networks have a different set of protocols
- It can be operated up-to application layer

References

1. William Stallings, "Data and Computer Communications", Seventh Edition, PHI 2004.
2. Andrew S. Tanenbaum, "Computer Networks" 4th Edition PHI
3. B. A. Fourozan, "TCP/IP Protocol Suite", 3rd Edition, Singapore, McGrawHill, 2004.
4. L. L. Peterson and B. S. Davie, Computer Networks-A System Approach, Elsevier.
5. B. A. Fourozan, "Data Communications and Networking", 4th Edition, Singapore, McGrawHill, 2004.
6. James F. Kurose, Keith W. Ross, "Computer Networking: A Top-Down Approach Featuring the Internet", 3rd Edition , Pearson Education 2009.
7. https://gaia.cs.umass.edu/kurose_ross/ppt.htm
8. PPT available for the respective books
9. <http://home.iitk.ac.in/~navi/sidbinetworkcourse/>

Thank You